Preprocessing steps Notes:

1)Binning groups similar values together, making patterns easier to detect.

Example: Age itself may not clearly show risk patterns, but grouping into "Young", "Middle-aged", and "Senior" may reveal trends in loan defaults by age group.

DrawBack:

1)When binning might reduce accuracy:

2)Loss of information

2) 2. Addressing Multicollinearity

✅ **Why use VIF?**

Identify redundant features:

If two or more features are highly correlated, you may want to remove or combine them.

Improve model reliability:

Reducing multicollinearity helps in getting more stable and interpretable coefficients.

Avoid misleading results:

Without checking VIF, you may incorrectly conclude that a feature is insignificant due to inflated standard errors.

Feature selection:

VIF helps in selecting the most important independent variables for your regression or machine learning model.

example:

You are building a loan default prediction model with these features:

Age

Income

LoanAmount

CreditScore

If LoanAmount and Income are highly correlated (people with higher income tend to borrow more), their VIF values might be high (>5 or 10), indicating multicollinearity.

3)Avoid Target Leakage

If you directly do TargetEncoder.fit_transform(df['LoanPurpose'], df['Default']) on the whole dataset, each row "sees" its own target value while being encoded.

Example: Suppose in the training data, all "Business" loans defaulted. The encoder would assign LoanPurpose="Business" → encoded value close to 1.0. This gives away information from the label, which is cheating.

4)Robust Scaling:

🔹 **What is RobustScaler?**

Standard StandardScaler uses:

$z=x-\mu/\sigma$ (subtract mean, divide by standard deviation).

RobustScaler instead uses:

$z=x-$median$/IQR$ where IQR = Q3 – Q1 (interquartile range).

👉 So instead of being influenced by extreme values (outliers), it uses statistics (median, IQR) that are more robust.

RobustScaler helps accuracy indirectly by stabilizing training and preventing outliers from dominating.

6) 🔹 What is Synthetic Feature Creation?

It means deriving new features from existing ones by combining them mathematically.

Examples:

Ratios: Income / LoanAmount

Products: MonthsEmployed * CreditScore

Differences: TotalDebt – Savings

These capture interactions between variables that a model might not automatically learn.

🔹 Why is it Useful for Loan Default Data?

In loan datasets, relationships between variables matter more than individual values.

Some examples:

Debt-to-Income Ratio (Income / LoanAmount)

A borrower earning 1,00,000 with a loan of 20,000 is safer than one earning 30,000 with a loan of 20,000.

Absolute numbers don’t tell the whole story, the ratio does.

✅ This is a strong risk predictor.

Usefulness: Captures financial stress and borrower stability.

7)With Pipeline + ColumnTransformer:

Automation → One object handles all preprocessing steps.

Example: Scale Income, RobustScale LoanAmount, TargetEncode LoanPurpose.

You don’t need to remember each step manually.

No Data Leakage → During cross-validation, the pipeline ensures transformations are fit only on training folds and applied to validation/test.

Consistency → The exact same preprocessing is applied when you deploy the model.

So the output is not directly shown, but internally you now have:

A fitted StandardScaler (means and stds of numeric columns).

A fitted OneHotEncoder (categories mapping).

A trained LogisticRegression model (coefficients, intercept).